Achalasia-Progeroid Syndrome (ACHPS): A Comprehensive Disease Characteristics Report

Disease: Achalasia-Progeroid Syndrome (ACHPS) **Identifiers:** OMIM #621123 · MONDO:0700300 · MedGen C6012702 (UID 1876522) · GARD 0027387 · UMLS C6012702 **Category:** Mendelian (ultra-rare, autosomal recessive spliceosomopathy) **Causal gene:** *BUD13* (HGNC:28199; NCBI Gene 84811; Ensembl ENSG00000137656; chromosome 11q23.3) **Prepared:** 2026-10-17

> **Evidence base & caveat.** ACHPS is an ultra-rare, newly delineated Mendelian disorder. Essentially all primary human clinical/molecular data derive from a **single defining study of five affected individuals** — Kornak et al., *Alternative splicing of BUD13 determines the severity of a developmental disorder with lipodystrophy and progeroid features* (PMID: 35670808). Mechanistic context is supplemented by zebrafish, yeast, and plant RES-complex studies. Clinical-management statements are, by necessity, extrapolated from the component phenotypes, as no disease-specific guidelines or trials exist. Where information is unavailable it is explicitly flagged.

Summary

Achalasia-progeroid syndrome (ACHPS) is an ultra-rare, autosomal-recessive, multisystem Mendelian disorder caused by biallelic loss-of-function variants in **BUD13**, a subunit of the spliceosomal **RES (REtention and Splicing) complex**. The disease was defined in 2022 by Kornak and colleagues, who identified five affected individuals — all carrying an identical homozygous nonsense variant, **NM_032725.4:c.688C>T, p.(Arg230Ter)** — through exome sequencing (PMID: 35670808). The clinical phenotype combines childhood-onset generalized lipodystrophy with a progeroid (prematurely aged) facial appearance, achalasia (failure of the lower esophageal sphincter to relax), corneal clouding, cataracts, retinal dystrophy,

progressive sensorineural hearing loss, demyelinating sensory neuropathy, and cardiac-valve regurgitation. In a severe subset, microcephaly, intellectual disability and death within the first decade of life also occur.

Mechanistically, BUD13 deficiency impairs pre-mRNA splicing. Patient-derived dermal fibroblasts show elevated intron retention, a global reduction of spliceosomal proteins, and nuclei with multiple nuclear-envelope invaginations. A striking feature is that the c.688C>T nonsense substitution does not simply trigger complete nonsense-mediated decay; instead it drives alternative splicing that produces a stable, truncated, **hypomorphic** BUD13 protein. The amount of this residual hypomorphic isoform positively correlates with milder disease and longer survival, explaining the strikingly **bimodal severity** observed across the five patients (three severe childhood-lethal cases versus two mildly affected adults with normal intellect).

There is no curative or gene-targeted therapy; management is entirely supportive and organ-directed (e.g., myotomy or pneumatic dilation for achalasia, cataract surgery, hearing aids, lipid-lowering therapy, cardiac surveillance). The disease is evolutionarily instructive: the RES complex is conserved from yeast through plants and zebrafish to humans, and **zebrafish bud13/RES-complex loss-of-function mutants** recapitulate a developmental neural phenotype, serving as the principal animal model. Because only five individuals have ever been reported, there are no population prevalence estimates, no registered clinical trials, and no Orphanet/ICD/MeSH codes assigned as of 2026.

Section 1 — Disease Information

Overview. ACHPS is a **spliceosomopathy** — a disorder of the pre-mRNA splicing machinery — presenting as a developmental disorder with lipodystrophy and progeroid features. It is defined by the combination of generalized lipodystrophy, a characteristic facial gestalt, and achalasia, together with sensory (ophthalmologic and auditory) and neurological involvement.

Key identifiers.

Resource	Identifier
OMIM	#621123
MONDO	MONDO:0700300
MedGen	C6012702 (UID 1876522)
GARD	0027387
UMLS	C6012702
Gene (HGNC)	*BUD13*, HGNC:28199
Gene (NCBI)	84811
Ensembl	ENSG00000137656
Orphanet / ICD-10 / ICD-11 / MeSH	No dedicated code assigned as of 2026

Synonyms / alternative names. Achalasia-progeroid syndrome; developmental disorder with lipodystrophy and progeroid features; BUD13-related lipodystrophy syndrome. The gene alias "ACHPS" is listed for *BUD13*.

Source of information. The disease-level knowledge is derived almost entirely from a single aggregated primary study (Kornak et al. 2022) describing five individuals, supplemented by aggregated ontology/database resources (OMIM, MedGen, ClinVar, UniProt, GO). It is therefore based on a small individual-patient case series rather than large EHR cohorts.

Section 2 — Etiology

Primary cause (genetic). ACHPS is a monogenic autosomal-recessive disorder. The cause is biallelic loss-of-function of *BUD13*. In the defining cohort, every affected individual carried the **identical homozygous nonsense variant** p.(Arg230Ter):

> *"All individuals harbored an identical homozygous nonsense variant affecting the retention and splicing complex component BUD13."* — PMID: 35670808

Genetic risk factors. The single causal determinant is homozygosity (or presumptively compound heterozygosity) for pathogenic *BUD13* loss-of-function alleles. The recurrence of one identical allele across all reported families is consistent with a **founder effect and/or consanguinity**. No susceptibility loci or validated modifier genes outside *BUD13* itself have been established.

Environmental risk factors. None identified. As a Mendelian disorder, environmental exposures are not causal. The only non-genetic contributors to outcome are supportive-care access and surgical management of complications.

Protective factors. No external protective factors are known. The key *intrinsic* modifier is molecular: the degree of residual **hypomorphic truncated BUD13 protein** produced by alternative splicing of the mutant allele acts as an endogenous protective modifier — higher residual activity correlates with milder disease and longer survival.

Gene–environment interactions. None demonstrated; the disorder is essentially fully genetically determined with an intrinsic (splicing-level) severity modifier rather than an environmental one.

Section 3 — Phenotypes

ACHPS is a multisystem disorder. The MedGen clinical synopsis encodes **26 HPO features**. Core features are summarized below with suggested HPO terms. The defining cohort reported variable severity, and some frequencies derive from the small n=5 series (so are qualitative).

> *"We identified a lipodystrophy phenotype with a typical facial appearance, corneal clouding, achalasia, progressive hearing loss, and variable severity."* — PMID: 35670808

Phenotype	HPO term	Type	Notes
Generalized lipodystrophy	HP:0009064	Physical manifestation	Childhood-onset; core feature
Excessive wrinkled skin	HP:0007392	Physical manifestation	Progeroid appearance
Dry skin	HP:0000958	Physical manifestation	—
Achalasia	HP:0002571	Clinical sign	Esophageal dysmotility
Corneal opacity (clouding)	HP:0007957	Clinical sign	—
Cataract	HP:0000518	Clinical sign	—
Retinal dystrophy	HP:0000556	Clinical sign	—
Sensorineural hearing loss	HP:0000407	Clinical sign	Progressive
Demyelinating sensory neuropathy	HP:0011402	Clinical sign / lab	—
Hypotonia	HP:0001252	Clinical sign	—
Intellectual disability	HP:0001249	Behavioral/ cognitive	Severe subset
Microcephaly	HP:0000252	Physical manifestation	Severe subset
Short stature	HP:0004322	Physical manifestation	—
Failure to thrive	HP:0001508	Clinical sign	—
Feeding difficulties	HP:0011968	Clinical sign	Related to achalasia
Triangular face	HP:0000325	Physical manifestation	Facial gestalt
Broad forehead	HP:0000337	Physical manifestation	Facial gestalt

Phenotype	HPO term	Type	Notes
Midface retrusion	HP:0011800	Physical manifestation	Facial gestalt
Deeply set eye	HP:0000490	Physical manifestation	Facial gestalt
Hypercholesterolemia	HP:0003124	Laboratory abnormality	—
Abnormal circulating lipid concentration	HP:0003119	Laboratory abnormality	—
Aortic regurgitation	HP:0001659	Clinical sign	—
Mitral regurgitation	HP:0001653	Clinical sign	—
Thickened aortic valve cusp	HP:0031568	Clinical sign	—
Joint contracture	HP:0034392	Physical manifestation	—
Short phalanx of finger	HP:0009803	Physical manifestation	—

Phenotype characteristics. - **Age of onset:** Childhood (some features congenital/infantile). Lipodystrophy and facial features appear early; hearing loss is progressive. - **Severity:** Variable/bimodal. In Kornak 2022 (n=5), three individuals (A1–A3) were severe and two (A4–A5) were mild. - **Progression:** Progressive for several features (hearing loss, neuropathy); multisystem accrual over time. - **Frequency:** From n=5 — severe phenotype in 3/5; adult survival with normal intellect in 2/5.

> *"3 individuals showed stunted growth, intellectual disability, and died within the first decade of life (A1, A2, and A3), 2 are adults with normal intellectual development (A4 and A5)"* — PMID: 35670808

Quality-of-life impact. Severe multisystem burden: feeding difficulties and achalasia impair nutrition; hearing and visual loss impair communication and independence; neuropathy and contractures impair mobility; in the severe subset, intellectual disability and early death are profound. No formal EQ-5D/SF-36 data exist for this ultra-rare disease.

Section 4 — Genetic / Molecular Information

Causal gene. *BUD13* (HGNC:28199; NCBI Gene 84811; Ensembl ENSG00000137656), located at **11q23.3**. Gene aliases: *ACHPS*, *Cwc26*, *fSAP71*. Protein: UniProt Q9BRD0, a **619-amino-acid** subunit of the spliceosomal RES complex.

Pathogenic variants.

Attribute	Detail
Defining variant	NM_032725.4:c.688C>T, **p.(Arg230Ter)**
ClinVar classification	Pathogenic (condition: Achalasia-progeroid syndrome)
Variant type	Nonsense (stop-gain)
Zygosity / origin	Homozygous, germline (recurrent allele)
Second listed variant	c.1693C>T p.(Arg565Cys) — conflicting classifications
Functional consequence	Loss of function converted to a **hypomorph** via alternative splicing

Population allele frequency and constraint. *BUD13* loss-of-function alleles are rare in gnomAD. Importantly, gnomAD constraint metrics show *BUD13* is **not** haploinsufficient: pLI ≈ 0 (1.6e-12); LoF observed/expected = 0.75 (90% CI 0.60–0.96); LoF Z = 1.66. This LoF tolerance is fully consistent with healthy heterozygous carriers and an **autosomal-recessive** disease mechanism.

Functional consequence — the splicing paradox. Rather than triggering complete nonsense-mediated decay, the p.Arg230Ter allele promotes alternative splicing that yields a stable, truncated, partially functional protein:

> *"The nucleotide substitution caused alternative splicing of BUD13 leading to a stable truncated protein whose expression positively correlated with disease expression and life expectancy."* — PMID: 35670808

Modifier genes. No external modifier genes identified. The principal modifier is intrinsic — the quantity of the hypomorphic BUD13 isoform.

Epigenetic information. No disease-specific DNA-methylation or histone-modification data are available for ACHPS.

Chromosomal abnormalities. None; ACHPS is a single-nucleotide (nonsense) disorder, not a structural/aneuploidy syndrome.

Section 5 — Environmental Information

- **Environmental factors:** None causal or contributory (monogenic disorder).
- **Lifestyle factors:** Not applicable to causation. Nutritional support is relevant to management but not etiology.
- **Infectious agents:** None. ACHPS is not infectious or trigger-associated.

Section 6 — Mechanism / Pathophysiology

Ordered causal chain

1. **Biallelic nonsense variant** *BUD13* c.688C>T p.(Arg230Ter) **→ leads to** loss of full-length BUD13 protein.
2. Loss of BUD13 **→ results in** a deficient/destabilized spliceosomal **RES complex** (BUD13–SNIP1/Pml1–RBMX2/Snu17) and **global reduction of spliceosomal proteins** (demonstrated in patient fibroblasts).
3. Impaired RES-complex function **→ leads to** defective pre-mRNA splicing, manifesting as **elevated intron retention** across many transcripts (demonstrated).
4. **Branch (severity modifier):** The mutant allele is **alternatively spliced** to a stable, truncated **hypomorphic** protein. *More* residual hypomorph **→ results in** milder disease / longer survival; *less* **→ results in** the severe, childhood-lethal phenotype (demonstrated: expression correlates with life expectancy).
5. Widespread mis-splicing/intron retention **→ leads to** altered expression of many downstream genes and **nuclear-envelope abnormalities** (nuclei with multiple invaginations; demonstrated, more pronounced in severe cases A1–A3).

6. Disrupted gene expression across tissues → **results in** the multisystem clinical phenotype — adipose (lipodystrophy), esophageal/enteric (achalasia), ocular (corneal clouding, cataract, retinal dystrophy), auditory (sensorineural hearing loss), peripheral nerve (demyelinating neuropathy), cardiac valves (regurgitation), brain (microcephaly, intellectual disability in severe subset), and skin (progeroid wrinkling) — *inferred* tissue-level consequences of the global splicing defect.
7. Rescue evidence: **overexpression of BUD13 isoforms normalized nuclear morphology** in patient cells (demonstrated), confirming BUD13 loss as the proximal driver.

```
BUD13 c.688C>T p.(Arg230Ter)  [homozygous, germline]
             | loss of full-length BUD13
             ▼
  Deficient RES complex  ──►  global ↓ spliceosomal proteins
             |
             ▼
  Defective pre-mRNA splicing ──► INTRON RETENTION (many genes)
             |                                  |
             |                                  ▼
  (BRANCH) alt. splicing of mutant      nuclear-envelope invaginations
  allele → stable truncated HYPOMORPH            |
             |  more hypomorph = milder          ▼
             ▼                              multisystem dysfunction
  severity / survival set here   ──►  lipodystrophy, achalasia, corneal
                                      clouding, cataract, retinal dystrophy,
                                      hearing loss, neuropathy, valve disease,
                                      microcephaly/ID (severe subset)
```

Supporting mechanistic detail

> *"In dermal fibroblasts, we found elevated intron retention, a global reduction of spliceosomal proteins, and nuclei with multiple invaginations, which were more pronounced in A1, A2, and A3."* — PMID: 35670808
>
> *"the alternative splice product converts a loss-of-function into a hypomorphic allele, thereby probably determining the sev[erity]..."* — PMID: 35670808

Molecular pathways / cellular processes. The core lesion is in **mRNA splicing via the spliceosome (GO:0000398)** and **U2-type prespliceosome assembly (GO:1903241)**. BUD13 functions in **RNA binding (GO:0003723)**. Downstream cellular processes (apoptosis/cell death, impaired neuronal differentiation) are inferred from the zebrafish model (see Section 15).

Protein dysfunction. BUD13 (UniProt Q9BRD0, 619 aa) is a RES-complex subunit; the truncating variant produces either loss of function (full NMD scenario) or a shortened hypomorph. RES is a heterotrimer with SNIP1 (Pml1) and RBMX2 (Snu17).

> *"The retention and splicing (RES) complex is formed by three different proteins (Bud13p, Pml1p and Snu17p) and is involved in splicing in yeast."* — PMID: 29969449

Metabolic changes. Secondary dyslipidemia (hypercholesterolemia; abnormal circulating lipids) accompanies the generalized lipodystrophy — a downstream metabolic consequence rather than a primary enzymatic defect.

Subcellular localization. Nucleus/nucleoplasm (GO:0005634 / GO:0005654); spliceosomal complex (GO:0005681); RES complex (GO:0070274); U2-type precatalytic spliceosome (GO:0071005); U2 snRNP (GO:0005686); also the minor/U12-type spliceosome (GO:0071018), indicating BUD13 participates in both major and minor splicing.

GO / CL term suggestions. Biological process: GO:0000398 (mRNA splicing, via spliceosome), GO:1903241 (U2-type prespliceosome assembly). Cellular component: GO:0070274 (RES complex), GO:0005681 (spliceosomal complex). Cell types involved: CL:0000136 (adipocyte), CL:0000057 (fibroblast — the in vitro model cell), CL:0002573 (Schwann cell — neuropathy), photoreceptor and enteric neuron populations (inferred).

Section 7 — Anatomical Structures Affected

Organ / system level (digestive, nervous, sensory, cardiovascular, integumentary, adipose, skeletal, craniofacial):

Structure	UBERON (suggested)	Manifestation
Adipose tissue	UBERON:0001013	Generalized lipodystrophy
Esophagus / lower esophageal sphincter	UBERON:0001043 / UBERON:0004550	Achalasia
Cornea	UBERON:0000964	Corneal clouding
Lens	UBERON:0000965	Cataract
Retina	UBERON:0000966	Retinal dystrophy
Inner ear / cochlea	UBERON:0001844	Sensorineural hearing loss
Peripheral nerve	UBERON:0001021	Demyelinating sensory neuropathy
Aortic valve	UBERON:0002137	Aortic regurgitation / thickened cusp
Mitral valve	UBERON:0002134	Mitral regurgitation
Brain	UBERON:0000955	Microcephaly, intellectual disability (severe subset)
Skin	UBERON:0002097	Wrinkled, dry (progeroid) skin

Tissue / cell level. Adipocytes (CL:0000136); Schwann cells (CL:0002573, demyelinating neuropathy); photoreceptors and enteric neurons (inferred); dermal fibroblasts (CL:0000057) are the principal in vitro model cell.

Subcellular level. Nucleus / nucleoplasm (GO:0005634 / GO:0005654) and the spliceosome / RES complex (GO:0005681 / GO:0070274). Patient fibroblasts characteristically show **nuclear-envelope invaginations**.

Lateralization. Involvement is systemic and bilateral/symmetric (e.g., bilateral ocular and auditory involvement).

Section 8 — Temporal Development

- **Onset:** Childhood (some features congenital/infantile) — insidious, chronic onset of a developmental disorder.

- **Progression:** Progressive and multisystem; features such as sensorineural hearing loss are explicitly progressive.
- **Disease course:** Chronic, lifelong. Bimodal trajectory — severe cases decline and die within the first decade; mild cases survive into adulthood.
- **Critical period / intervention window:** Early development (as suggested by the zebrafish model showing RES requirement during early embryogenesis). The severity-setting step (amount of hypomorphic isoform) is fixed at the molecular level, implying any splice-modulating intervention would ideally act early.
- **Remission:** None; no spontaneous or treatment-induced remission of the underlying disorder.

Section 9 — Inheritance and Population

Epidemiology. Ultra-rare. Only **5 affected individuals** have been reported worldwide (the Kornak 2022 defining cohort). No prevalence or incidence estimates exist.

Inheritance. Autosomal recessive (OMIM #621123; MONDO:0700300).

- **Penetrance:** Appears complete in biallelic carriers.
- **Expressivity:** Variable/bimodal, determined by the amount of hypomorphic truncated BUD13 isoform.
- **Genetic anticipation:** Not applicable (not a repeat-expansion disorder).
- **Germline mosaicism:** Not reported.
- **Founder effect / consanguinity:** Strongly suggested — all reported cases carry one identical recurrent homozygous allele.
- **Carrier frequency:** Unknown; *BUD13* LoF alleles are rare in gnomAD, and heterozygotes are healthy (gene is LoF-tolerant).

> *"All individuals harbored an identical homozygous nonsense variant affecting the retention and splicing complex component BUD13."* — PMID: 35670808

Population demographics. No established ethnic predilection beyond the founder allele; geographic distribution unknown; sex ratio not established (n too small); age distribution spans affected children (severe) and adults (mild).

Note on common *BUD13* variation. Common non-coding variants at the *APOA1/C3/A4/A5–ZPR1–BUD13* locus are associated with cardiometabolic (lipid) and cognitive traits in GWAS (e.g., PMID: 40665476). These are **distinct** from the rare biallelic coding LoF variants that cause ACHPS, but are mechanistically interesting given the lipid/dyslipidemia component of the syndrome.

Section 10 — Diagnostics

Primary diagnostic approach — molecular. Diagnosis is established by identifying **biallelic pathogenic *BUD13* variants** via **whole-exome sequencing** (the method used in Kornak 2022) or whole-genome sequencing. Once ACHPS is suspected clinically, single-gene or targeted-panel testing for *BUD13* is appropriate. Chromosomal microarray, karyotyping, FISH, mtDNA and repeat-expansion testing are **not** indicated (not a structural/mitochondrial/repeat disorder).

Organ-directed workup (mapped to phenotype):

Test	Target phenotype (HPO)
Esophageal manometry + barium swallow	Achalasia (HP:0002571)
Slit-lamp / ophthalmology exam	Corneal clouding (HP:0007957), cataract (HP:0000518), retinal dystrophy (HP:0000556)
Audiometry	Sensorineural hearing loss (HP:0000407)
Nerve conduction studies	Demyelinating sensory neuropathy (HP:0011402)
Echocardiography	Aortic/mitral regurgitation (HP:0001659 / HP:0001653)
Fasting lipid panel	Hypercholesterolemia (HP:0003124)

Research-grade / omics diagnostics. Dermal-fibroblast RNA-seq demonstrating intron retention; proteomics showing reduced spliceosomal proteins; electron microscopy showing nuclear-envelope invaginations. These are confirmatory functional assays rather than routine clinical tests.

Differential diagnosis — other congenital generalized lipodystrophy / progeroid disorders that can include achalasia:

Differential	Gene	Supporting evidence
CGL3	*CAV1*	Achalasia reported (PMID: 34643546)
CGL4	*CAVIN1/PTRF*	Achalasia reported (PMID: 40718185)
MDPL	*POLD1*	Progeroid lipodystrophy (PMID: 40718185)
Berardinelli–Seip (CGL1/2)	*AGPAT2*, *BSCL2*	Generalized lipodystrophy
Mandibuloacral dysplasia	*LMNA*, *ZMPSTE24*	Progeroid + lipodystrophy

> *"Two patients also presented at the age of 15 and 18 years with dysphagia due to achalasia"* — PMID: 34643546 (CAV1/CGL3)
>
> *"delayed puberty and achalasia cardia in the latter case"* — PMID: 40718185 (CAVIN1/CGL4)

Screening. No newborn or population carrier screening exists. Within affected families, **cascade carrier testing** and targeted testing for the familial *BUD13* allele are appropriate.

Section 11 — Outcome / Prognosis

Survival / mortality. Bimodal. In the defining cohort (n=5): three severe individuals (A1–A3) had stunted growth and intellectual disability and **died within the first decade**; two (A4–A5) are living adults with normal intellect.

Prognostic biomarker. The single most important prognostic determinant is molecular — the **expression level of the hypomorphic truncated BUD13 splice isoform**, which positively correlates with milder disease and longer life expectancy:

> *"the alternative splice product converts a loss-of-function into a hypomorphic allele, thereby probably determining the severity of the disease and the survival of affected individuals."* — PMID: 35670808

Morbidity. High across all survivors: feeding difficulty/failure to thrive, visual and auditory impairment, neuropathy, joint contractures, cardiac-valve disease, and dyslipidemia. No standardized QoL instruments have been applied.

Complications. Nutritional compromise (achalasia/feeding difficulty), aspiration risk, progressive sensory loss, cardiac valve dysfunction.

Recovery potential. None at the disease level; supportive interventions mitigate individual complications but do not reverse the disorder.

Section 12 — Treatment

No curative or gene-targeted therapy exists. Management is **supportive and organ-directed**. No clinical trials are registered (ultra-rare).

Problem (HPO)	Intervention	Suggested NCIT concept
Achalasia (HP:0002571)	Laparoscopic Heller myotomy; per-oral endoscopic myotomy (POEM); pneumatic balloon dilation; botulinum toxin injection	Esophagomyotomy; Esophageal Dilation
Feeding difficulty / FTT (HP:0011968 / HP:0001508)	Nutritional support; gastrostomy	Nutritional Support; Gastrostomy
Cataract (HP:0000518)	Cataract extraction	Cataract Surgery
Corneal clouding (HP:0007957)	Ophthalmologic management	—
Sensorineural hearing loss (HP:0000407)	Hearing aids; cochlear implant	Cochlear Implantation
Hypercholesterolemia (HP:0003124)	Lipid-lowering diet; statins	Statin Therapy
Valve regurgitation (HP:0001659 / HP:0001653)	Cardiology surveillance; valve repair if indicated	—
Neuropathy / contractures	Physical therapy, occupational therapy	Physical Therapy

Pharmacogenomics / advanced therapeutics. None established. The disease biology, however, points toward a rational experimental strategy: because residual hypomorphic BUD13 is protective, **splice-modulating therapy** (e.g., antisense oligonucleotides to bias splicing toward functional isoform, or read-through/stabilization approaches) is a biologically motivated but entirely hypothetical future direction.

Section 13 — Prevention

- **Primary prevention:** Not possible at the population level (monogenic, non-environmental). **Genetic counseling** for at-risk/consanguineous families is the principal preventive tool.
- **Reproductive options:** For couples with a known familial *BUD13* allele, carrier testing, prenatal diagnosis, and preimplantation genetic testing (PGT-M) are applicable.
- **Secondary prevention:** Early multidisciplinary surveillance (ophthalmology, audiology, cardiology, nutrition) to detect and manage complications promptly.
- **Tertiary prevention:** Organ-directed management to prevent complications (e.g., nutritional support to prevent aspiration/failure to thrive; valve surveillance).
- **Immunization / public-health / environmental interventions:** Not applicable.

Section 14 — Other Species / Natural Disease

- **Taxonomy of relevant models:** Zebrafish (*Danio rerio*, NCBI Taxon 7955); *Saccharomyces cerevisiae* (yeast); *Arabidopsis thaliana* (plant).
- **Orthologous genes / complex:** The RES complex is **evolutionarily conserved** — yeast Bud13/Snu17/Pml1; Arabidopsis AtBUD13/GDS1/DDL; zebrafish bud13/rbmx2/snip1; human BUD13/RBMX2/SNIP1.

> *"The heterotrimeric retention and splicing (RES) complex plays vital roles in the growth and development of yeast, zebrafish, and humans by mediating pre-mRNA splicing of multiple genes."* — PMID: 35640107

- **Naturally occurring disease in companion animals / wildlife:** No naturally occurring *BUD13*-related disease reported in OMIA for other species.
- **Comparative biology / conservation:** Deep conservation of the RES complex mechanism underscores that the splicing function is ancient and essential; the same molecular defect produces developmental phenotypes across phyla.
- **Zoonotic potential:** Not applicable (non-infectious genetic disorder).

Section 15 — Model Organisms

Zebrafish (principal disease model). Fernandez et al. 2018 generated loss-of-function mutants for all three RES-complex components (*bud13*, *snip1/pml1*, *rbmx2/snu17*), demonstrating they are required during early development and produce a **marked neural phenotype** with increased brain cell death and reduced differentiated neurons, alongside global intron-splicing defects:

> *"we have generated loss-of-function mutants for the three components of the RES complex in zebrafish and showed that they are required during early development. The mutants showed a marked neural phenotype with increased cell death in the brain and a decrease in differentiated neurons"* — PMID: 29969449

RES-dependent introns were found to be short, GC-rich, and intron-defined — consistent with the intron-retention phenotype seen in human ACHPS fibroblasts.

Other systems. - **Yeast (*S. cerevisiae*):** Defines the minimal RES heterotrimer (Bud13p/Snu17p/Pml1p); foundational for mechanism. - **Arabidopsis:** AtBUD13/GDS1/DDL RES orthologs regulate embryo and root development through splicing (PMID: 35640107). - **In vitro:** Patient-derived dermal fibroblasts (CL:0000057) are the key cellular model, reproducing intron retention, spliceosomal-protein reduction, and nuclear invaginations; rescue by BUD13 overexpression confirms causality.

Model characteristics. - *Phenotype recapitulation:* Zebrafish captures the neurodevelopmental/CNS aspect (brain cell death, neuronal deficit) and the molecular splicing defect well; patient fibroblasts recapitulate the cellular/nuclear phenotype. - *Limitations:* No reported mammalian (mouse) model of the specific hypomorphic human allele; zebrafish does not reproduce the full multisystem human gestalt (lipodystrophy, achalasia, progeroid features). A knock-in of the hypomorphic p.Arg230Ter allele would be needed to model the bimodal severity.

Resources: ZFIN (zebrafish), SGD (yeast), TAIR (Arabidopsis).

Mechanistic Model / Interpretation

ACHPS is best understood as a **dosage-sensitive spliceosomopathy with a built-in severity rheostat.** A single recurrent homozygous nonsense allele in *BUD13* would, at face value, be predicted to abolish protein via nonsense-mediated decay. Instead, the exact position of the premature stop codon triggers alternative splicing that salvages a stable, truncated, partially functional protein. The quantity of this salvaged hypomorph behaves like a dial: more residual BUD13 activity keeps splicing fidelity high enough for survival into adulthood with preserved cognition; less tips the system into severe global intron retention, spliceosome collapse, nuclear-envelope distortion, developmental failure, and death in the first decade. This elegantly explains how an **identical genotype** produces a **bimodal phenotype** — the variation is epi-transcriptomic (splicing-level), not a second genetic hit.

The downstream tissue pattern (adipose, esophagus, eye, ear, nerve, heart valve, brain, skin) reflects the broad reach of a general splicing defect combined with tissue-specific vulnerability. The convergence with other lipodystrophy-with-achalasia syndromes (CAV1, CAVIN1, POLD1) at the clinical level — despite distinct molecular causes — suggests shared downstream pathways affecting adipocyte biology and esophageal/enteric neuromuscular function that merit cross-disorder study.

Evidence Base

PMID	Title (abbrev.)	Role in this report
35670808	*Alternative splicing of BUD13 determines the severity of a developmental disorder with lipodystrophy and progeroid features*	**Defining study.** Establishes causal gene, recurrent homozygous nonsense allele, multisystem phenotype, bimodal severity, splicing mechanism (intron retention, spliceosomal-protein loss, nuclear invaginations), and hypomorphic-isoform prognostic model
29969449	*RES complex is associated with intron definition and required for zebrafish early embryogenesis*	Defines RES heterotrimer; principal **zebrafish model** with neural phenotype and intron-splicing defects
35640107	*An Arabidopsis Retention and Splicing complex regulates root and embryo development*	Establishes **evolutionary conservation** of RES across yeast, zebrafish, humans, plants
34643546	*Biallelic CAV1 null variants induce congenital generalized lipodystrophy with achalasia*	Key **differential diagnosis** (CGL3); documents achalasia in CAV1 lipodystrophy
40718185	*Inherited Lipodystrophy Associated With POLD1 and CAVIN1 Mutations*	**Differential diagnoses** (CGL4, MDPL); achalasia in CAVIN1 lipodystrophy
40665476	*Genome-wide study links cardiometabolic factors to cognition via APOA4-APOA5-ZPR1-BUD13...*	Context: common *BUD13*-locus variants and lipid/cognition traits (distinct from ACHPS coding LoF)

Evidence source types. Human clinical/genetic: P 35670808 (n=5 case series + functional fibroblast studies), 34643546, 40718185. Model organism / in vitro: P 29969449 (zebrafish), 35640107 (Arabidopsis). Population genetics / computational: gnomAD constraint

metrics, ClinVar classification, P 40665476 (GWAS).

Limitations and Knowledge Gaps

1. **Tiny evidence base (n=5).** All clinical knowledge derives from one cohort sharing a single recurrent allele. Phenotype frequencies, full spectrum, and penetrance/expressivity estimates are provisional.
2. **No epidemiology.** Prevalence, incidence, carrier frequency, geographic distribution, and sex ratio are all unknown.
3. **Single allele characterized.** Mechanism is defined for p.Arg230Ter only; whether other *BUD13* variants (e.g., the conflicting p.Arg565Cys) cause disease, and whether they produce the same hypomorphic-rescue behavior, is unestablished.
4. **No mammalian model of the human allele.** The zebrafish model captures CNS/splicing biology but not the full multisystem human phenotype (lipodystrophy, achalasia, progeroid features).
5. **No therapeutics or trials.** Management is supportive; splice-modulation is only a theoretical opportunity.
6. **Missing tissue-level mechanism.** Why specific tissues (esophagus, adipose, cornea) are vulnerable to a global splicing defect is not mechanistically resolved; the causal chain from mis-splicing to each organ phenotype is largely inferred.
7. **No epigenetic, metabolomic, or single-cell data** specific to ACHPS exist.
8. **No official Orphanet/ICD/MeSH codes** as of 2026, which limits registry-based ascertainment.

Proposed Follow-up Experiments / Actions

1. **Expand the patient registry via GeneMatcher/Matchmaker Exchange** to identify additional *BUD13* cases, define the full phenotype spectrum, establish frequencies, and test whether the founder allele is population-specific.

2. **Generate a knock-in mouse** carrying the hypomorphic p.Arg230Ter allele to model bimodal severity and the multisystem phenotype, and to quantify the hypomorph-dose/severity relationship in vivo.
3. **Characterize the hypomorphic isoform quantitatively** (patient fibroblast panels, RNA-seq + proteomics) to validate residual BUD13 level as a clinical prognostic biomarker, enabling genotype-independent severity prediction at diagnosis.
4. **Test splice-modulating ASOs** in patient fibroblasts to bias splicing toward the functional/hypomorphic isoform — a proof-of-concept for the rational therapeutic strategy implied by the disease biology.
5. **Map tissue-specific mis-splicing** (single-cell/organoid RNA-seq across adipose, esophageal/enteric, ocular, auditory lineages) to connect the global splicing defect to each organ phenotype.
6. **Functionally classify the conflicting c.1693C>T p.(Arg565Cys) variant** to clarify the ACHPS allelic series.
7. **Pursue formal disease-ontology coding** (Orphanet, ICD-11) to improve case ascertainment and knowledge-base integration.
